

Disaster Management System

By Remotz Systems

Final Deliverable

Project Team

Muhammad Moiz Ansari	23i-0523
Zohaib Umer	23i-0528
Rehan Tajik	23i-0730
Muhammad Tayyab	23i-0738

Semester 6

Supervised by
Ms. Rabail Zahid



**Department of Computer Science National University of
Computer and Emerging Sciences Islamabad, Pakistan**
April 29th, 2026

Table of Contents

PART A - Project Team Info	4
1. Team Introduction	4
2. Team Lead	4
3. Team Members	5
4. Role Assignments	6
5. Team Agreement	6
5.1 Methods of Communication	6
5.2 Communication Response Times	6
5.3 Meeting Attendance	6
5.4 Running Meetings	6
5.5 Meeting Preparation	6
5.6 Version Control	7
5.7 Division of Work	7
5.8 Submitting Assignments	7
5.9 Contingency Planning	7
PART B - Development Methodology (Scrum)	7
1. Sprint Backlog	7
1.1 Sprint Duration	7
1.2 Sprint Goal	7
1.3 Modules Developed	7
2. Sprint Backlog	8
2.1 Module Name	8
2.2 User Stories	8
2.3 Structured Specifications	23
3. Scrum Board (Trello Snapshots)	39
3.1 Snapshot 1 - Initial Backlog Upload	39
3.2 Snapshot 2 - Mid-Sprint Progress	39
3.3 Snapshot 3 - End of Sprint	39
4. Product Burn-Down Chart	40
5. Testing Activities	41
5.1. Software Test Plan	41
5.2. Black-Box Test Cases	42
5.3. White-Box Testing	44
5.4. Bug Report Summary	45
5.5. Test Results and Observations	46
PART C - Deliverable Document	47
1. Introduction	47
2. Project Vision	47

3. Intended Use of the System	47
4. Features / Overall Functionality	48
Sprint 1 - User Authentication & Disaster Reporting	48
Sprint 2 - Resource Management & Emergency Response	48
Sprint 3 - Coordination, Monitoring & System Completion	48
5. User Stories	49
6. Non-Functional Requirements (NFR)	50
7. Testing and Quality Assurance Summary	51
8. Implementation Screenshots	51
8.1 Login Page	51
8.2 Signup/Register Page	52
8.3 Manage Disaster Page	52
8.4 Manage Resources	53
8.5 NGO Dashboard	54
8.6 Rescue Team Dashboard	54
8.7 Assign NGO Page	55
8.8 Assign Rescue Team Page	56
8.9 Track Rescue Teams Page	56
8.10 Update Mission Status (Rescue Team Dashboard)	57
8.11 Generate Report Page	57
8.12 Give Feedback Page	58
8.13 Review Feedback Page (Admin)	58
9. Work Division	59

PART A - Project Team Info

1. Team Introduction

Team Name:

Remotz Systems

Team Logo:



Group Picture:



2. Team Lead

Team Lead Name	Moiz Ansari
Student ID	23i-0523

Role	Team Lead
-------------	-----------

3. Team Members

Member 1 — Moiz Ansari

Student ID: 23i-0523

GitHub Account: muhammad-moiz-ansari

Role(s): Scrum Master, UI Designer, Developer (Part-time)

Biography:

I am Moiz Ansari, a Computer Science student at FAST NUCES passionate about software development and system design. With a strong foundation in web development and UI design, my experience spans crafting custom game interfaces to engineering simple management systems like the TravelEase trip management database, focused on user-centric problem solving.



Member 2 — Zohaib Umer

Student ID: 23I-0528

GitHub Account: zohaib-umer

Role(s): Developer

Biography :

My name is Zohaib Umer. I am passionate about learning and exploring new ideas, especially in technology and computing. I have completed an internship, gaining practical experience and improving my skills. I enjoy problem-solving, reading, and working on projects that challenge me. I always strive to grow and learn.



Member 3 — Rehan Ali Tajik

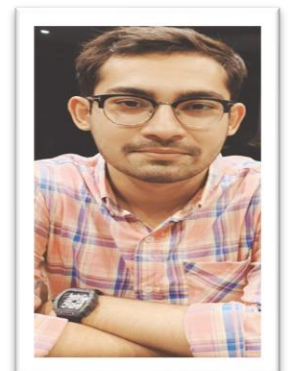
Student ID: 23I-0730

GitHub Account: RehanTajik730

Role(s): Developer, Requirement Analyst / Architect

Biography:

As a dedicated Computer Science student, I specialize in C++ and possess a strong foundation in software engineering principles. I am passionate about tackling complex programming challenges and optimizing system performance. I look forward to applying my analytical skills to develop an effective management system.



Member 4 — Hafiz Muhammad Tayyab

Student ID: 23i-0738

GitHub Account: TayyabMuhammad5

Role(s): Developer, Tester

Biography:

I am a motivated Computer Science student. I am passionate about software development, machine learning, and solving problems. I have practical experience in programming, using software development tools, and working on academic projects. I am excited to use my technical skills, keep up with new technologies, and contribute in team-based professional settings.



4. Role Assignments

Role	Assigned To (Name)
Product Owner / Project Manager	Moiz Ansari
Scrum Master	Moiz Ansari
Requirement Analyst / Architect	Rehan Tajik
Developer(s)	Moiz Ansari, Rehan Tajik, Zohaib Umer, Tayyab
Tester(s)	Tayyab
UI Designer	Moiz Ansari

5. Team Agreement

5.1 Methods of Communication

We will use our WhatsApp group as the primary source of quick communication, as it is mostly accessible and online for all users on multiple platforms and is used frequently. The tasks will be tracked on the Trello board. GitHub will be primarily used for code collaboration.

5.2 Communication Response Times

WhatsApp: Within 1.5 hours, Email: within 24 hours, Urgent matters: within 5 minutes

5.3 Meeting Attendance

The attendance of all group members in the meeting is necessary. The meeting will preferably be scheduled with suitable timings for all members. In case of any absentees, they will be informed about the discussion later.

5.4 Running Meetings

Online via Google Meet every Friday/Saturday at 6 PM, In-person meetings at the university on Thursdays between 1:00PM-2:30PM and in the mornings of Monday to Thursday. The Scrum Master will be responsible for taking meeting minutes and sharing them in the WhatsApp group.

5.5 Meeting Preparation

Assignment of members' tasks, issues faced in their tasks, planning of completion of work before the deadline.

5.6 Version Control

Each feature must be committed with proper, short, descriptive commit messages. Members must coordinate with each other what they are working on to avoid merge conflicts. Each member must keep a local copy of their work as well for backups. We will avoid committing IDE-specific files, compiled .class binaries, or local environment variables.

5.7 Division of Work

The Scrum Master assigns tasks during sprint planning based on member skills and availability. All members are equally responsible for code quality. The merging or report work will be mainly handled by the scrum master but all members will contribute equally. The work will be divided among Moiz, Rehan, Zohaib, and Tayyab based on individual strengths. The stakeholders would be primarily the citizens (Main Beneficiaries), NGOs, rescue teams, and the Admin.

5.8 Submitting Assignments

Final document must be reviewed by all members 24 hours before the due date; Team Lead submits on behalf of the group.

5.9 Contingency Planning

- (a) If a team member drops out: Redistribute workload and notify the instructor
 - (b) If a member consistently misses meetings: Formal warning then escalate to instructor
 - (c) If academic dishonesty: Zero tolerance, report to instructor immediately.
-

PART B - Development Methodology (Scrum)

1. Sprint Backlog

1.1 Sprint Duration: March 31, 2026 - April 18, 2026 (2.75 weeks)

1.2 Sprint Goal: Complete the disaster response coordination setup by implementing NGO/Rescue Team assignment, real-time mission tracking, status management, report generation, and feedback collection, delivering the fully integrated Disaster Management System.

1.3 Modules Developed:

Sprint 1:

Sprint 2:

Sprint 3: Coordination, Monitoring & System Completion

- Assign NGO (US-19 to US-21)
- Assign Rescue Team (US-22 to US-24)
- Track Rescue Team (US-25 to US-27)
- Update Mission Status (US-28 to US-30)

- Generate Disaster Report (US-31 to US-33)
- Give Feedback & Review Feedback (US-34, US-35)

2. Sprint Backlog

2.1 Module Name

Sprint 1: User Authentication & Disaster Reporting

Sprint 2: Resource Management & Emergency Response

Sprint 3: Coordination, Monitoring & System Completion

2.2 User Stories

Sprint 1:

User Story 1:

Story ID: US-01	Story Title: Register User	
User Story		Importance:
As a Citizen, NGO, or Rescue Team member, I want to register an account in the system so that I can access disaster management features according to my role		High
		Estimate: 3 days
Acceptance Criteria		Type:
• The system provides a registration form based on user role (Citizen / NGO / Rescue Team). • Users must enter required details (Name, CNIC / Registration No / Team ID, Contact Info, Password). • The system validates entered information. • If invalid data is entered, it shows an error message and asks the user to enter again. • Upon successful validation, the system creates the account and confirms registration. • System logs the registration action for auditing.		☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View

User Story 2:

Story ID: US-02	Story Title: Manage Disaster Records	
User Story		Importance:
As an Admin, I want to manage disaster records so that disaster information remains accurate, updated, and available for coordination and reporting.		High
		Estimate: 2 days
Acceptance Criteria		Type:

• The system displays a list of all disaster records. • Admin can choose to Add, Edit, or Delete a disaster record. • The system validates all changes before saving. • If a database error occurs, the system shows a failure message. • The system maintains logs of all updates for auditing.	☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View
--	---

User Story 3:

Story ID: US-03	Story Title: Add Disaster Record	
User Story	Importance:	
As an Admin, I want to add a new disaster record so that new disaster incidents are properly recorded and tracked in the system.	High	
	Estimate:	
	1 day	
Acceptance Criteria	Type:	
• The system provides a form to enter disaster details (type, location, severity, date, description). • The system validates required fields. • If invalid or incomplete data is entered, the system shows an error message. • Upon confirmation, the system stores the disaster record in the database. • The system logs the addition of the disaster record.	☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View	

User Story 4:

Story ID: US-04	Story Title: Edit Disaster Record	
User Story	Importance:	
As an Admin, I want to edit an existing disaster record so that I can update severity, status, or assigned teams when new information becomes available.	Medium	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
• The system allows the admin to select an existing disaster record. • The system displays editable disaster details. • The system validates updated information. • If invalid data is entered, the system shows an error message. • Upon confirmation, the system updates the disaster record in the database. • The system logs the modification with a timestamp.	☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View	

User Story 5:

Story ID: US-05	Story Title: Delete Disaster Record	
User Story		Importance:
As an Admin, I want to delete a disaster record so that incorrect or duplicate disaster entries can be removed from the system.		Medium
		Estimate: 2 days
Acceptance Criteria		Type:
• The system allows the admin to select a disaster record for deletion. • The system asks for confirmation before deleting. • If confirmed, the system removes the record from the database. • If deletion fails due to database error, the system shows a failure message. • System logs the deletion action for auditing purposes.		☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View

User Story 6:

Story ID: US-06	Story Title: Report Disaster	
User Story		Importance:
As an emergency responder or Citizen, I want to submit a disaster report so that the admin is immediately aware of the incident and a rescue operation is started.		High
		Estimate: 4 days
Acceptance Criteria		Type:
• System provides a form to select disaster type • Users can validate the entered information • If invalid data is entered it shows an error message and asks the user to enter again • Upon user confirmation the system records the disaster and notify authorities		☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View

User Story 7:

Story ID: US-07	Story Title: Send Notifications / Alert	
User Story		Importance:
As an admin I want to send notification/alert by providing necessary information such as location and severity so that the stakeholders are notified about the disaster.		High
		Estimate: 4 days
Acceptance Criteria		Type:
• System prompts the admin to specify type of disaster • If necessary information is missing the system asks the admin to complete them • The system forwards the details to the relevant stakeholders		☐ Search ☒ Workflow ☐ Manage Data ☐ Payment

	☐ Report / View
--	--

User Story 8:

Story ID: US-08	Story Title: Broadcast Alert to Stakeholders		
User Story		Importance:	
As a system I want to broadcast alerts to stakeholders so that the citizens and NGO's and the rescue teams receive notification to prepare and respond effectively.		High	
		Estimate:	
		2 days	
Acceptance Criteria		Type:	
- System connects to external communication channels to forward notifications. - Stakeholders must receive emergency alerts in 15 seconds - If alerts do not reach the system should resend it. - Critical alerts should be delivered even if system is in heavy load		☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 9:

Story ID: US-09	Story Title: Send Notifications		
User Story		Importance:	
As an admin, I want to send assignment notifications to NGO and rescue teams so that they know their task and begin their operations.		Medium	
		Estimate:	
		2 days	
Acceptance Criteria		Type:	
- System notifies the selected NGO or rescue team about the disaster assignment - The system allows the NGO or rescue team to acknowledge the assignment - Upon acknowledgement, the system must update disaster record and team status		☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 10:

Story ID: US-10	Story Title: View Disaster Alerts		
User Story		Importance:	
As a citizen, I want to view current disaster alerts so that I am aware of the current disaster events for my safety		High	
		Estimate:	
		1 day	
Acceptance Criteria		Type:	

• System retrieves and displays the list of current disaster alerts • The citizen can see the details of the disaster by clicking on it. • If no alert exists the system displays “No current disaster alerts”	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View
--	---

Sprint 2:

User Story 11:

Story ID: US-11	Story Title: Manage Resources		
User Story		Importance:	
As an Admin, I want to manage resources so that I can efficiently allocate, update, and track resources for disaster response operations.		High	
		Estimate:	
		3 day	
Acceptance Criteria		Type:	
• System provides options to Add, Edit, Delete, and Allocate resources. • System displays available resources with details (type, quantity, status). • System validates all inputs before saving changes. • If invalid data is entered, the system shows an error message. • System updates the resource database after confirmation. • System logs all resource management actions for auditing.		☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View	

User Story 12:

Story ID: US-12	Story Title: Add Resource		
User Story		Importance:	
As an Admin, I want to add new resources so that they can be available for allocation during disaster situations.		High	
		Estimate:	
		2 day	
Acceptance Criteria		Type:	
• System provides a form to enter resource details (type, quantity, description). • System validates required fields.		☐ Search ☐ Workflow ☒ Manage Data ☐ Payment	

• If invalid or incomplete data is entered, system shows an error message. • Upon confirmation, the system stores the new resource in the database. • System confirms successful addition of the resource. • System logs the addition action.	☐ Report / View
--	--

User Story 13:

Story ID: US-13	Story Title: Edit Resource		
User Story		Importance:	
As an Admin, I want to edit existing resources so that I can update their quantity, status, or details as needed		Medium	
		Estimate:	
		3 day	
Acceptance Criteria		Type:	
• System allows admin to select a resource to edit. • System displays current resource details. • Admin can modify resource information. • System validates updated data. • If invalid data is entered, system shows an error message. • Upon confirmation, system updates the resource in the database. • System logs the update action with timestamp.		☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View	

User Story 14:

Story ID: US-14	Story Title: Delete Resource		
User Story		Importance:	
As an Admin, I want to delete resources so that outdated or unnecessary resources can be removed from the system.		Medium	
		Estimate:	
		1 day	
Acceptance Criteria		Type:	
• System allows admin to select a resource for deletion. • System asks for confirmation before deleting. • If confirmed, system removes the resource from the database. • If deletion fails, system shows an error message. • System logs the deletion action for auditing purpose		☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View	

User Story 15:

Story ID: US-15	Story Title: Allocate Resource to NGO/Rescue Team	
User Story		Importance:
As an Admin, I want to allocate resources to an NGO or Rescue Team so that they have the necessary supplies to execute their disaster response operations.		High
		Estimate:
		2 days
Acceptance Criteria		Type:
• The system allows the Admin to select a specific NGO or Rescue Team from an active list. • The system provides an interface to select the resource type and quantity to be assigned. • The system verifies availability and deducts allocated items from the central inventory. • The system displays a success message confirming the allocation and notifies the team.		☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View

User Story 16:

Story ID: US-16	Story Title: View Allocated Resources	
User Story		Importance:
As an NGO / Rescue Team member, I want to view the resources allocated to my team so that I know what equipment and supplies are available for our current mission.		High
		Estimate:
		1 day
Acceptance Criteria		Type:
• System retrieves and displays a list of resources currently allocated to the logged-in user's team. • The user can view specific details such as resource type, quantity, and date of allocation.		☐ Search ☐ Workflow ☐ Manage Data

If no resources are currently assigned, the system displays "No allocated resources found."	☐ Payment ☒ Report / View
---	---

User Story 17:

Story ID: US-17	Story Title: Request Additional Resources	
User Story	Importance:	
As an NGO / Rescue Team member, I want to request additional resources so that we can swiftly replenish our supplies and avoid shortages during a disaster response.	Medium	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
- User can access a request form and select the required resource type and desired quantity. - User must provide a mandatory brief justification or reason for the request. - Upon submission, the system automatically routes the request to an Admin for review. - System displays "Request submitted successfully" and updates the request status to 'Pending'.	☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 18:

Story ID: US-18	Story Title: Call For Emergency	
User Story	Importance:	
As a citizen, I want to trigger an emergency call so that I can receive immediate, location-based assistance during a life-threatening disaster.	Critical	
	Estimate:	
	3 days	
Acceptance Criteria	Type:	
- Citizen can activate the emergency alert via a prominent, single-action SOS button. - System automatically captures the user's current GPS location and attaches it to the alert. - System immediately routes the distress signal to the nearest available Rescue Teams and Admin. - System displays an immediate confirmation message to the	☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

citizen indicating help is dispatched.	
--	--

Sprint 3:

User Story 19:

Story ID: US-19	Story Title: Assign NGO		
User Story		Importance:	
As an Admin, I want to assign NGOs to disaster-affected locations so that I can dispatch an appropriate NGO efficiently.		High	
		Estimate:	
		3 day	
Acceptance Criteria		Type:	
- The system provides options to track the availability of NGOs - System validates that only available NGOs are assigned. - If an unavailable NGO is assigned, the system shows an error message. - System updates the NGO database after confirmation. - System logs all NGO assignment actions.		☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 20:

Story ID: US-20	Story Title: View Available NGOs		
User Story		Importance:	
As an Admin, I want to be able to view all available NGOs so I can make decisions on which NGO is to be assigned to a specific mission.		Medium	
		Estimate:	
		1 day	
Acceptance Criteria		Type:	
- The system provides a tabular display listing NGOs, their information(location, personnel, speciality) and availability. - The system updates the status of availability as soon as the NGO is assigned.		☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 21:

Story ID: US-21	Story Title: Assign Rescue Team	
User Story	Importance:	
As an Admin, I want to assign Rescue Teams to disaster-affected locations so that I can dispatch an appropriate Rescue Team efficiently.	High	
	Estimate:	
	3 day	
Acceptance Criteria	Type:	
• The system provides options to track the availability of Rescue Teams • System validates that only available Rescue Teams are assigned. • If an unavailable Rescue Team is assigned, the system shows an error message. • System updates the Rescue Team database after confirmation. • System logs all Rescue Team assignment actions.	☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 23:

Story ID: US-23	Story Title: View Available Rescue Teams	
User Story	Importance:	
As an Admin, I want to view available rescue teams so that I can select the most suitable team for disaster response.	Medium	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
• System displays a list of all rescue teams. • System shows team details (location, availability, specialization). • System filters only available teams. • If no teams are available, system shows an appropriate message. • Data is retrieved in real-time.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 24:

Story ID: US-24	Story Title: Assign Team to Disaster
-----------------	--------------------------------------

User Story	Importance:
As an Admin, I want to assign a rescue team to a disaster so that rescue operations can start quickly and efficiently.	High
	Estimate:
	3 days
Acceptance Criteria	Type:
• System displays disasters requiring rescue teams. • System shows available rescue teams. • Admin selects a team for a disaster. • System validates availability of selected team. • Upon confirmation, system assigns the team. • System notifies the rescue team. • System updates disaster and team status.	☐ Search ☒ Workflow ☒ Manage Data ☐ Payment ☐ Report / View

User Story 25:

Story ID: US-25	Story Title: Track Rescue Team	
User Story	Importance:	
As an Admin, I want to track rescue teams so that I can monitor their progress and ensure timely response.	High	
	Estimate:	
	4 days	
Acceptance Criteria	Type:	
• System provides tracking option for rescue teams. • System retrieves real-time team data. • Admin can select a specific team to track. • System displays current mission details. • If team is offline, system shows alert message.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 26:

Story ID: US-26	Story Title: View Active Teams Dashboard	
User Story	Importance:	
	Medium	

As an Admin, I want to view an active teams dashboard so that I can get an overview of all ongoing rescue operations.	Estimate:
	2 days
Acceptance Criteria	Type:
• System displays all active rescue teams. • Dashboard shows assigned disasters and status. • Data updates in real-time. • If no teams are active, system displays a message.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View

User Story 27:

Story ID: US-27	Story Title: View Team Location & Status	
User Story	Importance:	
As an Admin, I want to view team location and status so that I can monitor progress and coordinate effectively.	High	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
• System shows real-time location of selected team. • System displays mission status (assigned, in progress, completed). • System updates data dynamically. • If location is unavailable, system shows an alert.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 28:

Story ID: US-28	Story Title: Update Mission Status	
User Story	Importance:	
As an NGO or Rescue Team member, I want to update mission status so that the admin stays informed about progress.	High	
	Estimate:	
	3 days	

Acceptance Criteria	Type:
• System allows NGO/Rescue Team to update mission status. • System provides status options (In Progress, Completed). • System validates the update. • System records changes in database. • System notifies admin of updates.	☐ Search ☒ Workflow ☒ Manage Data ☐ Payment ☐ Report / View

User Story 29:

Story ID: US-29	Story Title: Mark Mission In Progress	
User Story	Importance:	
As an Admin, I want to view available rescue teams so that I can select the most suitable team for disaster response.	Medium	
	Estimate:	
	1 days	
Acceptance Criteria	Type:	
• System allows selection of assigned mission. • User selects “In Progress” status. • System validates the action. • System updates mission status in database. • Admin is notified of the update.	☐ Search ☒ Workflow ☐ Manage Data ☐ Payment ☐ Report / View	

User Story 30:

Story ID: US-30	Story Title: Mark Mission Completed	
User Story	Importance:	
As an Admin, I want to view available rescue teams so that I can select the most suitable team for disaster response.	Medium	
	Estimate:	
	1 days	
Acceptance Criteria	Type:	
• System allows selection of assigned mission. • User selects “Completed” status.	☐ Search ☒ Workflow	

• System validates completion. • System updates mission status in database. • Admin is notified of completion. • System logs mission completion.	☐ Manage Data ☐ Payment ☐ Report / View
---	--

User Story: 31

Story ID: US-31	Story Title: Generate Disaster Report	
User Story		Importance:
As an Admin, I want to generate detailed reports so that I can review disaster impact, resources used, and team efficiency.		High
		Estimate:
		34 days
Acceptance Criteria		Type:
• The system must compile disaster events, affected areas, response teams, relief resources distributed, and outcomes. • The system must log the report generation action for auditing purposes. • If report generation fails due to a system error, the admin must be prompted to retry.		☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View

User Story 32:

Story ID: US-32	Story Title: Set Report Parameters	
User Story		Importance:
As an admin, I want to set specific parameters for the report so that the system can organize disaster data matching my criteria.		Medium
		Estimate:
		3 days
Acceptance Criteria		Type:

• System must prompt the admin for report parameters like date, disaster type, region, and teams involved. • Admin must be able to set desired parameters for the report into the system. • If no disaster match the entered parameters, the system must display the exact message: "No disaster data found for selected parameters".	☐ Search ☐ Workflow ☒ Manage Data ☐ Payment ☐ Report / View
---	---

User Story 33:

Story ID: US-33	Story Title: Export/Download Report	
User Story	Importance:	
As an Admin, I want to select a report format so that I can view or download the generated report.	Medium	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
• System must prompt the admin to request a specific report format. • System must successfully generate the report in the chosen format, supporting options like PDF or Excel. • Admin must be able to view or download the report	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ☒ Report / View	

User Story 34:

Story ID: US-34	Story Title: Give Feedback	
User Story	Importance:	
As a citizen, NGO, or rescue team member, I want to share experiences or feedback for the system's performance so that the system could be improved.	Low	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
• System must provide a feedback form once the	☐ Search	

authenticated actor selects the option to give feedback. • If the feedback form is incomplete, the user must be prompted to fill in the mandatory fields. • If a database error occurs during submission, the system must show a failure message.	■ Workflow ☐ Manage Data ☐ Payment ☐ Report / View
---	--

User Story 35:

Story ID: US-35	Story Title: Review Feedback	
User Story	Importance:	
As an admin, I want to view the submitted feedback so that I can review the feedback for the system's performance.	Low	
	Estimate:	
	2 days	
Acceptance Criteria	Type:	
• System must store all recorded feedback properly. • All recorded feedback must be accessible for review by the admin through the interface.	☐ Search ☐ Workflow ☐ Manage Data ☐ Payment ■ Report / View	

2.3 Structured Specifications

Sprint 1:

Specification 1 - US-: Manage Disaster Records

Use Case Section	Comment												
Use Case Name	Manage Disaster Records												
Scope	Disaster Management System												
Level	User-Goal												
Primary Actor	Admin												
Stakeholders and Interests	Admin: Wants to create, update, delete and maintain accurate disaster records. Users/NGOs/Rescue Teams: Depend on accurate data for effective planning and response in the future.												
Main Success Scenario	<table> <tr> <th>Actor Action</th><th>System Response</th></tr> <tr> <td>1. Admin selects the option to manage disaster records.</td><td></td></tr> <tr> <td></td><td>2. The system displays a list of disaster records.</td></tr> <tr> <td>3. Admin adds new records/modifies existing records.</td><td></td></tr> <tr> <td></td><td>4. System saves/validates records.</td></tr> <tr> <td></td><td>5. Prompts admin upon status of completion.</td></tr> </table>	Actor Action	System Response	1. Admin selects the option to manage disaster records.			2. The system displays a list of disaster records.	3. Admin adds new records/modifies existing records.			4. System saves/validates records.		5. Prompts admin upon status of completion.
Actor Action	System Response												
1. Admin selects the option to manage disaster records.													
	2. The system displays a list of disaster records.												
3. Admin adds new records/modifies existing records.													
	4. System saves/validates records.												
	5. Prompts admin upon status of completion.												
Extensions	• If a disaster record does not exist, the system notifies the admin • If data is in the wrong format, the system prompts the admin for correct data. • If database error occurs, the system notifies the admin.												
Pre-conditions	• Admin is authorized • Database is connected												
Post-conditions	• Disaster records are successfully recorded. • Logs maintained for future reporting.												

Specification 2 - US-10: View Disaster Alerts

Use Case Section	Comment																		
Use Case Name	View Disaster Alerts																		
Scope	Disaster Management System																		
Level	User-Goal																		
Primary Actor	Citizen , Admin																		
Stakeholders and Interests	Citizen: Wants to stay informed about current disaster events for safety. Admin: Needs visibility of all reported disasters for effective coordination. Rescue Team/NGO: May use alerts to prepare for response.																		
Main Success Scenario	<table border="1"> <thead> <tr> <th>Actor Action</th><th>System Response</th></tr> </thead> <tbody> <tr> <td>1. The actor selects the option to view disaster alerts.</td><td></td></tr> <tr> <td></td><td>2. System retrieves and displays the list of current disaster alerts</td></tr> <tr> <td>3. The actor selects a specific disaster alert.</td><td></td></tr> <tr> <td></td><td>4. The system shows details: location, severity, status, and response teams assigned.</td></tr> <tr> <td></td><td></td></tr> <tr> <td></td><td></td></tr> <tr> <td></td><td></td></tr> <tr> <td></td><td></td></tr> </tbody> </table>	Actor Action	System Response	1. The actor selects the option to view disaster alerts.			2. System retrieves and displays the list of current disaster alerts	3. The actor selects a specific disaster alert.			4. The system shows details: location, severity, status, and response teams assigned.								
Actor Action	System Response																		
1. The actor selects the option to view disaster alerts.																			
	2. System retrieves and displays the list of current disaster alerts																		
3. The actor selects a specific disaster alert.																			
	4. The system shows details: location, severity, status, and response teams assigned.																		
Extensions	• If no alerts exist, the system displays: <i>"No current disaster alerts."</i> • If network/ data error occurs, system prompts user to retry																		
Pre-conditions	• Disasters must be reported in the system. • Citizen must have access to system application.																		
Post-conditions	• Citizen/Admin successfully views the disaster alerts.																		

Specification 3 - US-07: Send Notification/Alert

Use Case Section	Comment														
Use Case Name	Send Notification/Alert														
Scope	Disaster Management System														
Level	User-Goal														
Primary Actor	Admin														
Stakeholders and Interests	Admin: Wants to efficiently and quickly notify stakeholders about a disaster. NGO/Rescue Team: Needs timely alerts to prepare and respond effectively. Citizens: Expect to receive disaster alerts to take safety precautions as early as possible.														
Main Success Scenario	<table> <tr> <th>Actor Action</th><th>System Response</th></tr> <tr> <td>1. Admin decides to send a notification/alert.</td><td></td></tr> <tr> <td></td><td>2. The system prompts the admin to specify the type of disaster alert.</td></tr> <tr> <td>3. Admin provides necessary information such as location of disaster, its severity and instructions.</td><td></td></tr> <tr> <td></td><td>4. The system accepts/validates information.</td></tr> <tr> <td></td><td>5. System forwards alert to all registered stakeholders.</td></tr> <tr> <td>6. Stakeholders receive the notification in real-time.</td><td></td></tr> </table>	Actor Action	System Response	1. Admin decides to send a notification/alert.			2. The system prompts the admin to specify the type of disaster alert.	3. Admin provides necessary information such as location of disaster, its severity and instructions.			4. The system accepts/validates information.		5. System forwards alert to all registered stakeholders.	6. Stakeholders receive the notification in real-time.	
Actor Action	System Response														
1. Admin decides to send a notification/alert.															
	2. The system prompts the admin to specify the type of disaster alert.														
3. Admin provides necessary information such as location of disaster, its severity and instructions.															
	4. The system accepts/validates information.														
	5. System forwards alert to all registered stakeholders.														
6. Stakeholders receive the notification in real-time.															
Extensions	• If necessary information is missing, the system prompts the admin to complete them. • If alerts don't reach some users, the system resends the alert.														
Pre-conditions	• Admin is authorized. • System is connected to communication channels (														
Post-conditions	Notifications are forwarded to all relevant stakeholders.														

Specification 4 - US-06: Report Disaster

Use Case Section	Comment																
Use Case Name	Report Disaster																
Scope	Disaster Management System																
Level	User-Goal																
Primary Actor	Citizen																
Stakeholders and Interests	● Citizen: Wants to let admin know to receive rescue help and resources timely. ● Admin: Wants to get informed about disaster to start taking actions.																
Main Success Scenario	<table> <tr> <th>Actor Action</th><th>System Response</th></tr> <tr> <td>1. The user selects the option Report Disaster.</td><td></td></tr> <tr> <td></td><td>2. The system shows the form input.</td></tr> <tr> <td>3. The user enters the required details.</td><td></td></tr> <tr> <td></td><td>4. The system validates the entered information.</td></tr> <tr> <td>5. User confirms to report the disaster.</td><td></td></tr> <tr> <td></td><td>6. The system notifies the admin.</td></tr> <tr> <td></td><td>7. System updates the reporting log.</td></tr> </table>	Actor Action	System Response	1. The user selects the option Report Disaster.			2. The system shows the form input.	3. The user enters the required details.			4. The system validates the entered information.	5. User confirms to report the disaster.			6. The system notifies the admin.		7. System updates the reporting log.
Actor Action	System Response																
1. The user selects the option Report Disaster.																	
	2. The system shows the form input.																
3. The user enters the required details.																	
	4. The system validates the entered information.																
5. User confirms to report the disaster.																	
	6. The system notifies the admin.																
	7. System updates the reporting log.																
Extensions	4. Invalid data entered - Shows an error message “Invalid data entered”, and prompts the user to enter data again. 6. Notification failure - The data is stored on the system temporarily until the admin is notified about it. 7. Log failure - System shows “Update failed. Try again later.”																
Pre-conditions	● The user has access to the system application. ● The system is online and connected to disaster database.																
Post-conditions	● Disaster report is recorded in the database. ● Authorities are notified of the new disaster report. ● Rescue operations may be initiated based on the report on admin’s command.																

Sprint 2:

Specification 5 - US-11: Manage Resources

Field	Details
Use Case Name	Manage Resources
Actor(s)	Admin
Inputs	• Admin chooses an option to add, remove, update, or allocate resources. • Enters required details in the form. • Confirms allocation.
Outputs	• System shows management options and input forms. • Displays validation/error messages. • Shows update status.
Processing	• System validates the entered information. • Assigns resources to the Rescue team or NGO. • Updates the database.

Preconditions	• Admin must be logged in to an account. • Resources must be available. • Rescue team or NGO must be registered.
Postconditions	• The resource database is updated. • Allocated resources are reflected in team/NGO accounts. • Logs of resource actions are stored.
Exceptions	• Invalid data entered. • Insufficient resources. • Database error.

Specification 6 - US-16: View Allocated Resources

Field	Details
Use Case Name	View Allocated Resources
Actor(s)	NGO

Inputs	NGO selects the option to view allocated resources.
Outputs	System shows a database of allocated resources.
Processing	System retrieves and displays the allocated resource records.
Preconditions	• NGO must be registered and authorized • Resources must be allocated by the admin and available in the database.
Postconditions	NGO can plan and assign resources to its teams (Medical, transport, etc.).
Exceptions	Database Disconnected (System shows error message "Can't load database" and attempts to load data again).

Specification 7 - US-17: Request Additional Resources

Field	Details
Use Case Name	Request Additional Resources
Actor(s)	NGO
Inputs	NGO requests more resources if needed.
Outputs	System responds with a resource update.
Processing	System processes the request and updates resources (if the admin approves).
Preconditions	NGO must be registered and authorized.
Postconditions	Resource allocation is updated for the NGO to distribute.
Exceptions	Response Error (System sends response again).

Specification 8 - US-18: Call for emergency

Field	Details
Use Case Name	Call for Emergency
Actor(s)	Citizen
Inputs	• Citizen selects "Call for Emergency" • Provides necessary information (location and emergency type) • Sends confirmation.
Outputs	• System alerts the nearest rescue team. • Notifies the admin • Updates the citizen about rescue team progress.
Processing	• System validates information. • Routes the alert to the closest available rescue team. • Notifies the admin. • Updates progress upon team acknowledgment.
Preconditions	• User must have access to the system application. • System is connected to emergency contact networks and rescue databases.

Postconditions	• Emergency alert is sent • System records the emergency call in the database • Admin is notified for follow-up actions.
Exceptions	• Citizen fails to provide location. • Invalid data entered. • Notification failure. • Rescue team unavailable.

Sprint 3:

Specification 9-US-19: Assign NGO, Rescue Team

Field	Details
Use Case Name	Assign NGO / Assign Rescue Team
Actor(s)	Admin
Inputs	• Admin selects an option to assign a team or NGO. • Selects a specific disaster. • Selects a relevant and available rescue team or NGO for the assignment.

Outputs	• System displays lists of disasters requiring assistance. • Shows available NGOs/Rescue Teams. • Notifies the assigned entity.
Processing	• System filters available teams based on location, availability, and skill • Filters NGOs based on relevance and availability • Assigns the team and updates disaster records and team status upon acknowledgement.
Preconditions	• Disaster events exist in the system • NGOs and rescue teams are registered and available.
Postconditions	• NGO or Rescue Team is assigned to the disaster incident • The assigned entity acknowledges the assignment.
Exceptions	• If no entities are available, the system alerts the admin • If admin cancels, the assignment is discarded • Admin can reschedule or reassign if the first rescue team cannot respond.

Specification 10-US-25: Track Rescue Team

Field	Details

Use Case Name	Track Rescue Team
Actor(s)	Admin
Inputs	• Admin selects options to track rescue teams • Selects a specific rescue team.
Outputs	• System retrieves a list of all active rescue teams and their assignments • Displays the team's current location, disaster assignment, status, and estimated arrival time.
Processing	System updates real-time location and status based on inputs from the rescue team.
Preconditions	• Rescue teams are registered in the system • Teams are assigned to at least one disaster.
Postconditions	Admin successfully views the real-time status of rescue teams.
Exceptions	• If a rescue team is offline/unreachable, the system alerts the admin

	• If no teams are active, the system displays "No active rescue teams currently".
--	---

Specification 11-US-31: Generate Disaster Report

Field	Details
Use Case Name	Generate Disaster Report
Actor(s)	Admin
Inputs	• Admin selects option to generate disaster reports • Enters report parameters (date, disaster type, region, teams involved) • Requests report format.
Outputs	• System prompts for parameters • Generates reports in the chosen format • Displays or downloads the report.

Processing	• System compiles disaster events, affected areas, response teams, relief resources distributed, and outcomes • Logs report generation for auditing purposes.
Preconditions	Disaster events, rescue assignments, and resource records exist in the system.
Postconditions	Admin successfully generates and views/downloads the disaster reports.
Exceptions	• If no disasters match the parameters, the system displays "No disaster data found for selected criteria." • If report generation fails due to system error, admin is prompted to retry.

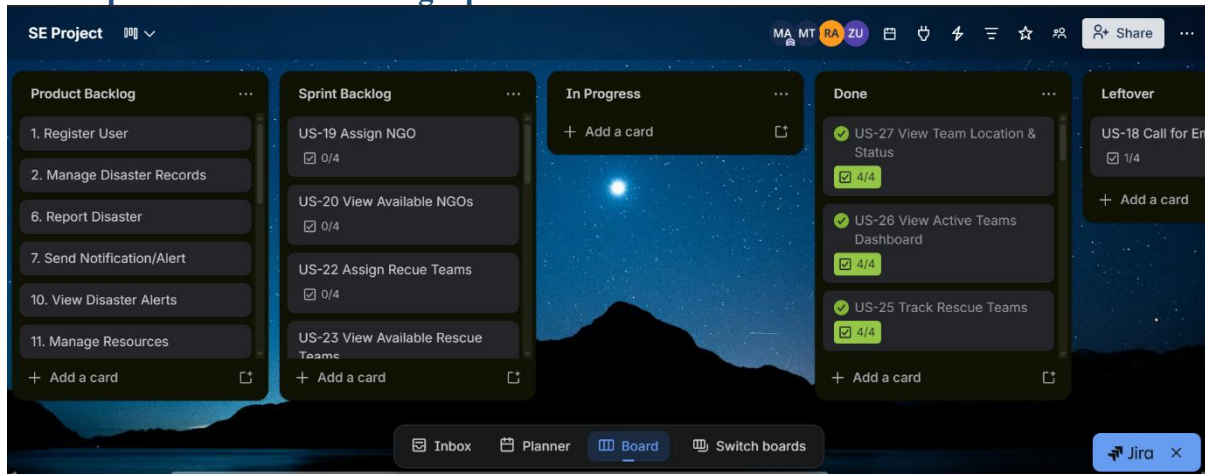
Specification 12-US-34: Give Feedback

Field	Details

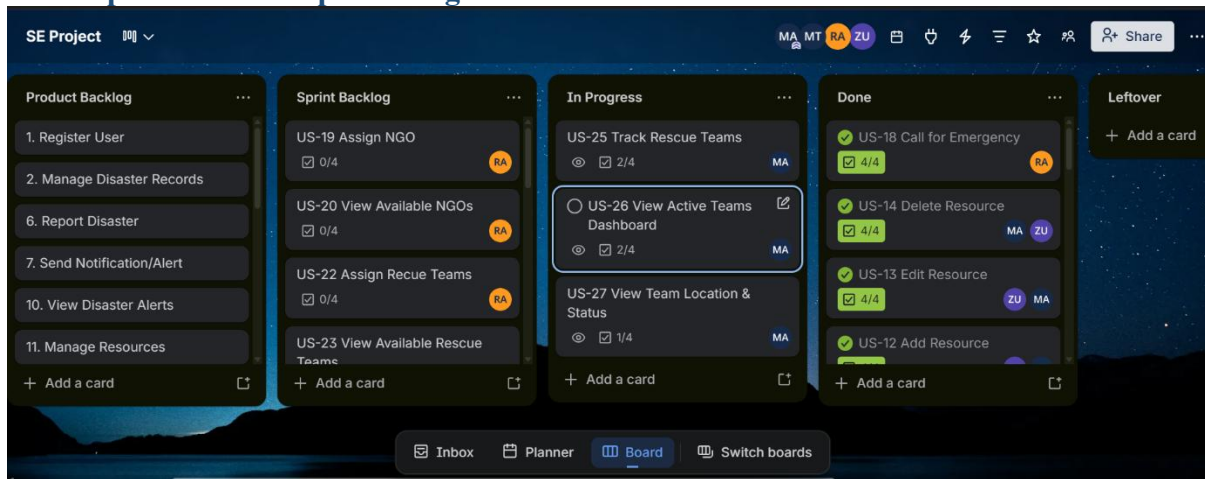
Use Case Name	Give Feedback
Actor(s)	Citizen, NGO, Rescue Team
Inputs	• Actor selects the option to give feedback • Actor enters feedback into the provided form.
Outputs	System provides a feedback form.
Processing	• System records the submitted feedback • Admins can later review the feedback.
Preconditions	The actor is authenticated.
Postconditions	• Feedback is recorded • Respective feedback is accessible to admin, NGO, rescue teams.
Exceptions	• If the feedback form is incomplete, the user is prompted to fill mandatory fields • If a database error occurs, the system shows a failure message.

3. Scrum Board (Trello Snapshots)

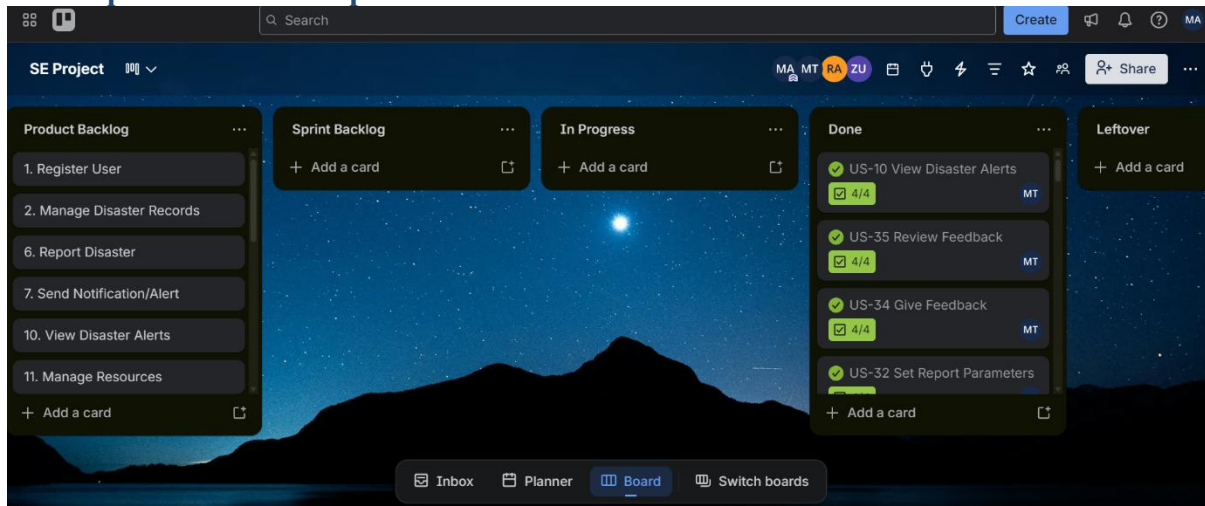
3.1 Snapshot 1 - Initial Backlog Upload



3.2 Snapshot 2 - Mid-Sprint Progress

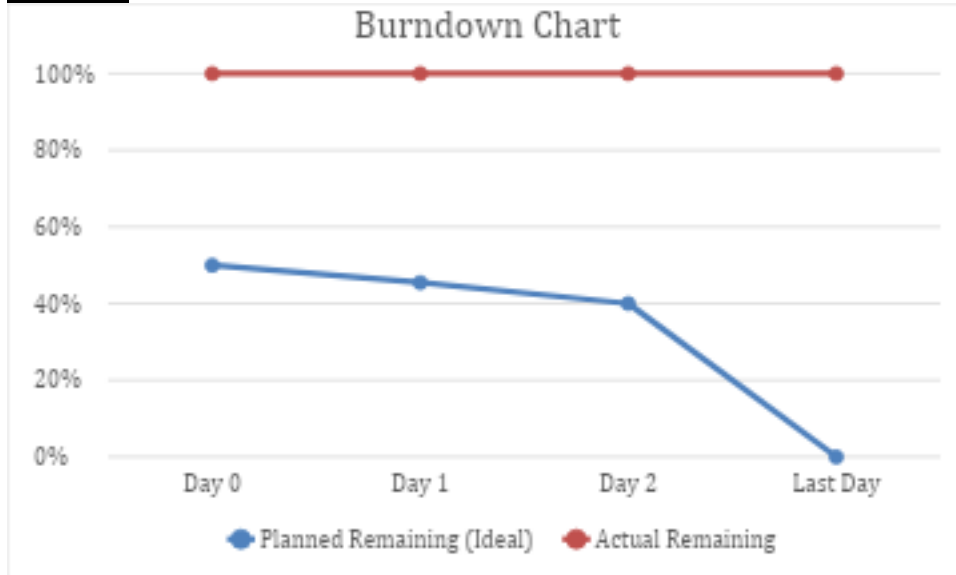


3.3 Snapshot 3 - End of Sprint

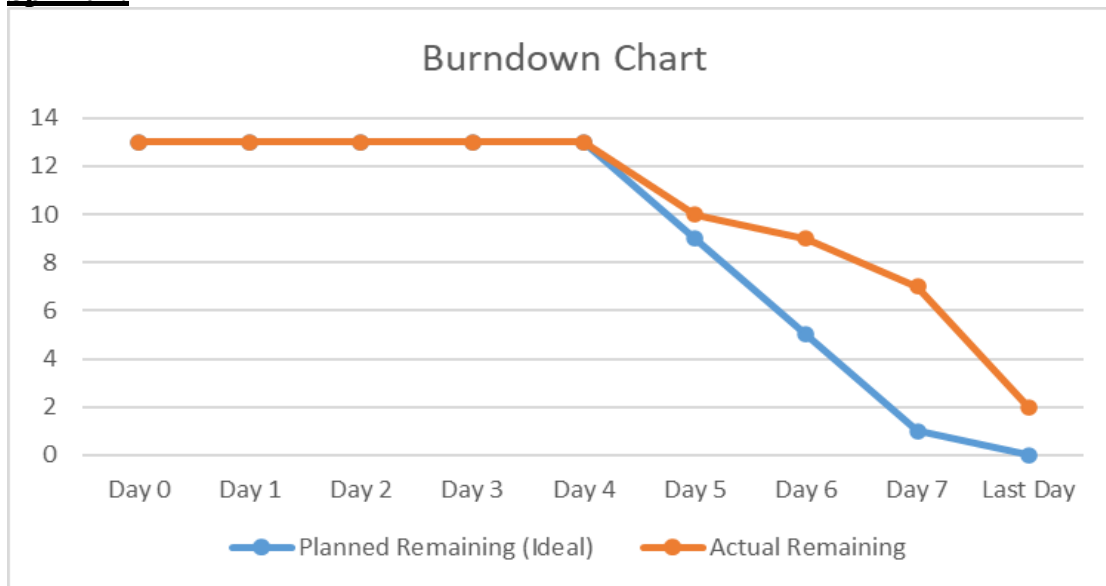


4. Product Burn-Down Chart

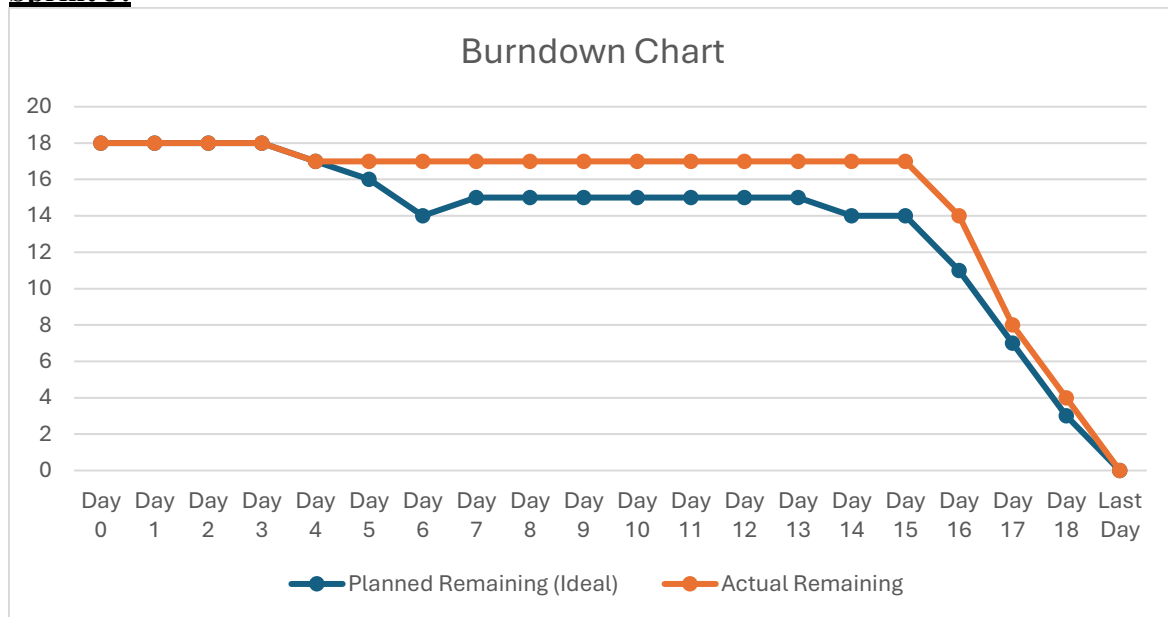
Sprint 1:



Sprint 2:



Sprint 3:



5. Testing Activities

5.1. Software Test Plan

Test Plan Title	Sprint 3 Software Test Plan - Disaster Management System
Version	1.0
Prepared By	Remozone Systems — Muhammad Moiz Ansari, Rehan Tajik, Zohaib Umer, Muhammad Tayyab
Sprint	Sprint 3 — Coordination, Monitoring & System Completion
Test Period	April 1, 2026 – April 18, 2026
Scope	This test plan covers all Sprint 3 modules: • Assign NGO (US-19 to US-21) • Assign Rescue Team (US-22 to US-24) • Track Rescue Team (US-25 to US-27) • Update Mission Status (US-28 to US-30) • Generate Disaster Report (US-31 to US-33) • Give & Review Feedback (US-34, US-35)

Test Objectives	1. Verify that NGO and Rescue Team assignment correctly updates all related database records. 2. Verify that mission status transitions are enforced and reflected in real time. 3. Verify that report generation produces correct data for all filter combinations. 4. Verify that feedback submission and review workflows function correctly. 5. Verify that all exception paths display appropriate error messages without crashing.
Test Types	• Functional Testing — each use case against its acceptance criteria • Black-Box Testing — equivalence partitioning and boundary value analysis • White-Box Testing — statement, branch, and condition coverage on critical methods • Regression Testing — ensure Sprint 1 & Sprint 2 features remain unaffected
Tools & Environment	IDE: IntelliJ IDEA / Java 17 Database: SQL Server Express (local) Manual testing via JavaFX UI Test evidence: screenshots and console output logs
Entry Criteria	• Sprint 3 features are fully coded and compiled without errors. • Database schema matches the expected Sprint 3 design. • Sprint 1 & 2 features pass regression checks.
Exit Criteria	• All critical and high-priority test cases pass. • No open Critical or High severity defects remain unfixed. • All modules produce correct outputs for valid inputs. • Exception paths display user-friendly error messages.

5.2. Black-Box Test Cases

Techniques used: Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Error Guessing (EG)

TC-ID	Feature / Use Case	Input(s)	Expected Output	Technique	Result
TC-01	Assign NGO	Valid: Active disaster + Available NGO selected	NGO assigned; disaster record updated; success message	Equivalence Partitioning	Pass
TC-02	Assign NGO	Invalid: No NGOs available in system	Alert: 'No NGOs currently available.' Assignment blocked	Error Guessing	Pass

TC-03	Assign NGO	Invalid: Disaster already has NGO assigned	System prevents duplicate assignment	Equivalence Partitioning	Pass
TC-04	Assign Rescue Team	Valid: Active disaster + Available team selected	Team assigned; Mission record created; team status = Deployed	Equivalence Partitioning	Pass
TC-05	Assign Rescue Team	Invalid: Selected team has status 'Deployed'	Team card greyed out; cannot be selected	Equivalence Partitioning	Pass
TC-06	Track Rescue Team	Valid: Active missions exist in system	Teams list populated with correct status badges and ETAs	Equivalence Partitioning	Pass
TC-07	Track Rescue Team	Edge: No active missions	Empty state displayed: 'No active rescue teams currently.'	Boundary Value Analysis	Pass
TC-08	Update Mission Status	Valid: Change status from Assigned → On Site	Status updated in DB; timeline entry added; Admin dashboard refreshed	Equivalence Partitioning	Pass
TC-09	Update Mission Status	Edge: Status already Completed — attempt to change	System prevents change; status options for completed missions are disabled	Error Guessing	Pass
TC-10	Generate Report	Valid: Date range Jan 1 – Dec 31, All types, All severities	Report generated with correct totals; export buttons enabled	Equivalence Partitioning	Pass
TC-11	Generate Report	Invalid: Start date > End date	Error shown: 'Invalid date range.' Generation blocked	Boundary Value Analysis	Pass
TC-12	Generate Report	Edge: No disasters match the filter criteria	'No disaster data found for selected criteria.' Preview shows zeros	Boundary Value Analysis	Pass
TC-13	Give Feedback	Valid: All fields filled, 4-star rating, non-anonymous	Feedback saved with status 'Pending'; confirmation shown	Equivalence Partitioning	Pass
TC-14	Give Feedback	Invalid: Rating not selected (0 stars)	Validation error: 'Please rate your experience (1–5 stars).'	Boundary Value Analysis	Pass
TC-15	Give Feedback	Invalid: Feedback type = NGO Service but no entity selected	Validation error: 'Please select a specific NGO for this feedback type.'	Error Guessing	Pass

TC-16	Give Feedback	Invalid: Non-anonymous but email field left empty	Validation error: 'Please enter your email address.'	Equivalence Partitioning	Pass
TC-17	Review Feedback	Valid: Admin marks feedback as Resolved	Feedback status updated; counter updated on Admin dashboard	Equivalence Partitioning	Pass
TC-18	Review Feedback	Valid: Admin searches for feedback by keyword	Filtered list shows only matching feedback cards	Equivalence Partitioning	Pass

5.3. White-Box Testing

Method Under Test	performAssignment() — AssignRescueTeamController
Description	Assigns a rescue team to a disaster, creates a Mission, and updates team/disaster status.
Coverage Criteria	Branch Coverage
Paths/Conditions Tested	Path 1 (Happy): teamUpdated=true AND disasterUpdated=true AND missionCreated=true → shows success, reloads data. Path 2 (Team update fail): teamStatusUpdated=false → shows error, no mission created. Path 3 (Disaster update fail): assignSuccess=false → shows error, rollback attempted. Path 4 (Mission creation fail): missionCreated=false → partial success warning shown.
Coverage Result	4 of 4 branches covered (100% branch coverage)

Method Under Test	validateForm() — GiveFeedbackController
Description	Validates all required fields before feedback submission.
Coverage Criteria	Condition Coverage

Paths/Conditions Tested	Condition 1: feedbackTypeCombo.getValue() == null → returns false, shows error. Condition 2: Feedback type requires entity AND entityCombo = N/A → returns false. Condition 3: currentRating == 0 → returns false, shows star rating error. Condition 4: subjectField is empty → returns false. Condition 5: feedbackArea is empty → returns false. Condition 6: !anonymous AND email invalid → returns false. Condition 7: All conditions pass → returns true, submission proceeds.
Coverage Result	7 of 7 conditions tested (100% condition coverage)

Method Under Test	updatePreview() — GenerateReportController
Description	Fetches live data and updates the report preview panel based on selected filters.
Coverage Criteria	Statement Coverage
Paths/Conditions Tested	Statement 1: startDate or endDate is null → sets error title, resets labels, returns early. Statement 2: startDate.isAfter(endDate) → same early return path as Statement 1. Statement 3: Valid date range → fetches totals, updates all preview labels. Statement 4: selectedSeverity != 'All Severities' → appends severity tag to preview title. Statement 5: breakdown is empty → sets 'No disaster data found' in breakdown container.
Coverage Result	All 5 statement branches executed (100% statement coverage)

5.4. Bug Report Summary

Bug ID	Module	Severity	Description	Status	Fix Applied
BUG-01	Assign Rescue Team	High	Mission ID generator duplicated IDs when multiple missions existed with non-sequential IDs (e.g., MIS001, MIS010 → generated MIS002 instead of MIS011).	Fixed	Replaced manual ID generator with DB query fetching MAX ID.

BUG-02	Track Rescue Team	Medium	Map pin did not appear when team's city was not found in MAPLOCATION table; NullPointerException thrown.	Fixed	Added null check in updatePinPosition(); falls back to default map center coordinates.
BUG-03	Manage Disasters	High	The status of rescue and ngo team didn't update when the disaster was deleted.	Fixed	Added team id fetcher and updated teams' statuses, if they were assigned to the respective disaster..
BUG-04	Review Feedback	Low	The color of labels on feedback differed from the buttons of labels.	Fixed	Swapped the colors of labels.
BUG-05	Review Feedback	Low	Average rating label on Admin Dashboard showed 'NaN/5.0' when no feedback existed in the database.	Fixed	Added null/NaN guard in loadFeedbackStats(); defaults to 0.0 when no ratings exist.

5.5. Test Results and Observations

Total test cases executed: 18

Passed: 18 | Failed: 0 | Blocked: 0

Key observations:

- All critical user flows (assign NGO, assign rescue team, mission status update) pass with correct database state transitions.
- Report generation correctly handles boundary cases such as equal start/end dates and empty result sets.
- Feedback validation correctly blocks submission for all identified invalid input combinations.
- Five bugs were identified during testing; all five were resolved before sprint completion.
- White-box analysis confirmed 100% branch, condition, and statement coverage on the three critical methods tested.

How testing improved system quality:

Testing in Sprint 3 specifically uncovered the Mission ID duplication bug, which could have caused data integrity failures in production. The null-pointer exception in the Track Rescue Teams map view was also caught before delivery, preventing a runtime crash. Systematic condition coverage of the feedback validator eliminated all edge-case submission bypasses. Overall, the test activities increased confidence in the correctness and robustness of the final integrated system.

PART C - Deliverable Document

1. Introduction

The Disaster Management System (DMS) is a desktop-based Java application developed to streamline the coordination, communication, and resource management required during natural disasters such as floods, earthquakes, and fires in Pakistan. The system serves four user roles — Admin, Rescue Team, NGO, and Citizen — each with a dedicated dashboard and role-specific functionality.

Sprint 1 established the foundational layer of the system: user registration and authentication, disaster record management, disaster alert broadcasting, and citizen disaster reporting.

Sprint 2 extended the system with a centralized resource inventory, dynamic resource allocation to response teams and NGOs, field resource request capabilities, and a citizen emergency SOS feature.

With Sprint 3, the system has reached full operational completion. The final sprint implements the entire response coordination loop: assigning NGOs and rescue teams to disasters, tracking deployed team locations and mission timelines, enabling field teams to update their mission status in real time, generating comprehensive disaster reports, and capturing post-operation feedback. Together, the three sprints deliver a complete, integrated Disaster Management System ready for operational use.

2. Project Vision

To create a centralized, resilient platform that empowers communities and response organizations to collaborate effectively during disasters, ultimately saving lives and reducing property damage through timely information and organized action.

With the completion of Sprint 3, this vision is fully realized. The DMS now covers the entire disaster lifecycle — from initial citizen reporting through coordinated response, real-time tracking, post-operation reporting, and performance feedback. The system ensures that no disaster goes unaddressed due to communication failures, resource misallocation, or lack of coordination between response entities.

3. Intended Use of the System

Stakeholders and Updated Needs

Stakeholder	Updated Needs
Citizens	Need to report disasters, request emergency help, view active alerts, and provide post-operation feedback on response quality.
NGOs	Need to view disaster assignments, manage relief distribution, update mission status, request additional resources, and submit performance feedback.

Rescue Teams	Need to receive mission assignments, update their deployment status in real time (En Route → On Site → Completed), track resources allocated to them, and submit post-mission feedback.
Admin	Needs full control of disaster lifecycle management: record creation, alert broadcasting, NGO/team assignment, resource allocation, team tracking, report generation, and feedback review.
Government Agencies	Need access to generated disaster reports (PDF/Excel/CSV) for policy decisions, budget planning, and operational accountability.

4. Features / Overall Functionality

Sprint 1 - User Authentication & Disaster Reporting

- Secure role-based User Authentication and Registration (Admin, Citizen, NGO, Rescue Team).
- Citizen-driven Disaster Reporting and Admin-driven Disaster Record Management (CRUD).
- Real-time Alert Broadcasting to all registered stakeholders.
- Citizen alert viewing dashboard.

Sprint 2 - Resource Management & Emergency Response

- **Centralized Resource Inventory:** Admins can add, edit, delete, and track master supplies.
- **Dynamic Resource Allocation:** Admins assign specific quantities of resources to NGOs and rescue teams.
- **Field Resource Requests:** NGOs and Rescue Teams can view allocations and request further assistance.
- **Emergency SOS:** A dedicated high-priority emergency trigger for citizens in immediate danger.

Sprint 3 - Coordination, Monitoring & System Completion

- **NGO Assignment:** Admin assigns available NGOs to active disasters; NGO notified automatically.
- **Rescue Team Assignment:** Admin assigns available teams; Mission record created automatically.
- **Real-Time Team Tracking:** Admin monitors deployed team locations on a Pakistan map with status timelines.
- **Mission Status Management:** Rescue Teams and NGOs update mission status (Assigned → En Route → On Site → Completed).
- **Disaster Report Generation:** Admin generates filterable reports and exports to PDF, Excel, or CSV.
- **Feedback System:** Citizens, NGOs, and Rescue Teams submit rated feedback; Admin reviews and resolves.

How Sprint 3 Completes System Capability

Sprint 3 closes the disaster response loop that Sprints 1 and 2 initiated. Without Sprint 3, the system could detect and respond to a disaster, but had no mechanism to formally deploy response entities, monitor their progress, or evaluate the quality of the response. The addition of the NGO/Team assignment creates the formal deployment chain. Track Rescue Team provides the Admin with operational visibility. Update Mission Status gives field teams a voice in the system. Generate Report transforms raw data into actionable intelligence. Give/Review Feedback creates a continuous improvement mechanism. Together, these features elevate the DMS from a data entry tool to a fully operational disaster coordination platform.

5. User Stories

The following user stories were implemented. Full backlog tables are provided in Part B, Section 2.

Sprint 1:

- **US-01:** Register User
- **US-02:** Manage Disaster Records (Epic)
 - **US-03:** Add Disaster Record
 - **US-04:** Edit Disaster Record
 - **US-05:** Delete Disaster Record
- **US-06:** Report Disaster
- **US-07:** Send Notification/Alert (Epic)
 - **US-08:** Broadcast Alert to Stakeholders
 - **US-09:** Send Notification
- **US-10:** View Disaster Alerts

Sprint 2:

- **US-11:** Manage Resources (Admin)
 - **US-12:** Add Resource (Admin)
 - **US-13:** Edit Resource (Admin)
 - **US-14:** Delete Resource (Admin)
 - **US-15:** Allocate Resource to NGO/Rescue Team (Admin)
- **US-16:** View Allocated Resources (NGO, Rescue Team)
- **US-17:** Request Additional Resources (NGO, Rescue Team)

US-18: Call for Emergency (Citizen)

Sprint 3:

- **Assign NGO**
 - **US-19:** Assign NGO (Admin)

- US-20: View Available NGOs (Admin)
- **Assign Rescue Team**
 - US-22: Assign Rescue Team (Admin)
 - US-23: View Available Rescue Teams (Admin)
 - US-24: Assign Team to Disaster (Admin)
- **Track Rescue Team**
 - US-25: Track Rescue Team (Admin)
 - US-26: View Active Teams Dashboard (Admin)
 - US-27: View Team Location & Status (Admin)
- **Update Mission Status**
 - US-28: Update Mission Status (NGO, Rescue Team)
 - US-29: Mark Mission In Progress (NGO, Rescue Team)
 - US-30: Mark Mission Completed (NGO, Rescue Team)
- **Generate Disaster Report**
 - US-31: Generate Disaster Report (Admin)
 - US-32: Set Report Parameters (Admin)
 - US-33: Export/Download Report (Admin)
- **Feedback**
 - US-34: Give Feedback (Citizen, NGO, Rescue Team)
 - US-35: Review Feedback (Admin)

6. Non-Functional Requirements (NFR)

NFR ID	Category	Natural Language Specification
NFR-01	Performance	The system shall load the Admin Dashboard within 3 seconds under normal database conditions with up to 1,000 records across all tables.
NFR-02	Performance	Alert broadcast to all registered stakeholders shall be completed within 5 seconds of Admin confirmation.
NFR-03	Availability	The system shall be available 99.5% of the time during operational hours to ensure citizens can report disasters at any time.
NFR-04	Security	Only users with the Admin role shall have access to the Manage Disaster Records, Broadcast Alert, Assign NGO, Assign Rescue Team, and Generate Report modules.
NFR-05	Security	All user passwords shall be stored in the database using irreversible hashing; plaintext passwords shall never be stored or transmitted.
NFR-06	Usability	The Report Disaster form shall be navigable and submittable within 3 user interactions to ensure ease of use during emergencies.

NFR-07	Usability	New users shall be able to complete their primary task (report a disaster, submit feedback, update mission status) within 5 minutes without prior training.
NFR-08	Scalability	The system shall support at least 5,000 concurrent users accessing different dashboards without performance degradation.
NFR-09	Scalability	The Report Generation module shall handle datasets of up to 10,000 disaster records without exceeding a 15-second generation time.
NFR-10	Reliability	The system shall handle all invalid user inputs gracefully, displaying appropriate error messages without crashing or corrupting data.
NFR-11	Data Integrity	All assignment operations (NGO/Rescue Team) shall be performed within database transactions; partial failures shall trigger automatic rollback.
NFR-12	Maintainability	The system shall follow a layered architecture (Controller → Service → DAO), ensuring each layer is independently testable and modifiable.

7. Testing and Quality Assurance Summary

Full testing artifacts are provided in Part B, Section 6. The following summarizes the testing approach and results for Sprint 3.

- **Test Plan:** Covers all 6 Sprint 3 modules with defined objectives, entry/exit criteria, and test types (functional, black-box, white-box, regression).
- **Black-Box Testing:** 18 test cases designed using Equivalence Partitioning, Boundary Value Analysis, and Error Guessing. All 18 passed.
- **White-Box Testing:** Statement, branch, and condition coverage applied to 3 critical methods — `performAssignment()`, `validateForm()`, and `updatePreview()`. 100% coverage achieved on all three.
- **Bug Report:** 5 bugs identified and resolved — including a critical Mission ID duplication bug and a `NullPointerException` in the map tracking module.
- **Quality Impact:** Testing improved system reliability, prevented data integrity failures, eliminated runtime crashes, and validated all user-facing validation messages.

8. Implementation Screenshots

Sprint 1:

8.1 Login Page

Disaster Management System - Login



Disaster Management System

Coordinating Response. Saving Lives.

Emergency Alert
*** Report disasters immediately to enable rapid response

Welcome Back

Sign in to your account to continue

Sign In

Sign Up

Login as

Admin

Username

Enter your username

Password

Enter your password

Forgot password?

Emergency Hotline: **1122**

8.2 Signup/Register Page

Disaster Management System - Login



Disaster Management System

Coordinating Response. Saving Lives.

Emergency Alert
*** Report disasters immediately to enable rapid response

Welcome Back

Sign in to your account to continue

Sign In

Sign Up

Register as

Citizen

Name *

Enter your full name

CNIC *

Enter CNIC (e.g., 12345-1234567-1)

Contact Number *

Enter phone number

Emergency Hotline: **1122**

8.3 Manage Disaster Page

DMS - Manage Resources

Disaster Management SystemAdmin UserLogout

MENU

- Dashboard
- Send Alert
- Manage Disasters
- Assign NGO
- Assign Rescue Team
- Track Rescue Teams
- Manage Resources
- Generate Report
- Review Feedback

Manage Resources

Track inventory and allocate disaster relief resources

Inventory

Requested Resources

Filter by Status:

All

Pending

Allocated

Rejected

ID	Resource	Quantity	Requester	Priority	Status	Requested Date	Action
PKG001	Medical Supplies Kit (15 Un...	15 Units	Pakistan Red Cresc...	High	Allocated	Nov 22, 2025 05:5...	
PKG002	Emergency Tents (10 Units)	10 Units	Alpha Rescue Squad	Critical	Rejected	Nov 22, 2025 05:5...	
PKG003	First Aid Kits (20 Units)	20 Units	Red Crescent Karachi	Medium	Allocated	Nov 22, 2025 05:5...	
PKG004	Food Packages (50 Boxes)	50 Boxes	Beta Rescue Team	High	Allocated	Nov 22, 2025 05:5...	
PKG005	Blankets (50 Units)	50 Units	Pakistan Red Cresc...	High	Allocated	Nov 22, 2025 06:2...	
PKG006	Food Packages (100 Boxes)	100 Boxes	RT005	Critical	Allocated	Nov 22, 2025 07:4...	
PKG007	Emergency Tents (50 Units)	50 Units	RT006	Critical	Rejected	Nov 22, 2025 08:2...	

8.5 NGO Dashboard

DMS - NGO Dashboard

Disaster Management SystemPakistan Red CrescentLogout

MENU

- Dashboard
- View Alerts
- Give Feedback

NGO Dashboard

Red Crescent Pakistan - Relief Operations Overview

Active Assignme...
1

Resources Distribut...
150

Beneficiaries Serv...
500

Pending Reque...
3

Current Assignments

Flood

Rawal Dam Area, Islamabad

Resolved

Details

Earthquake

Swat Valley

Under Control

Details

Recent Distribution Activities

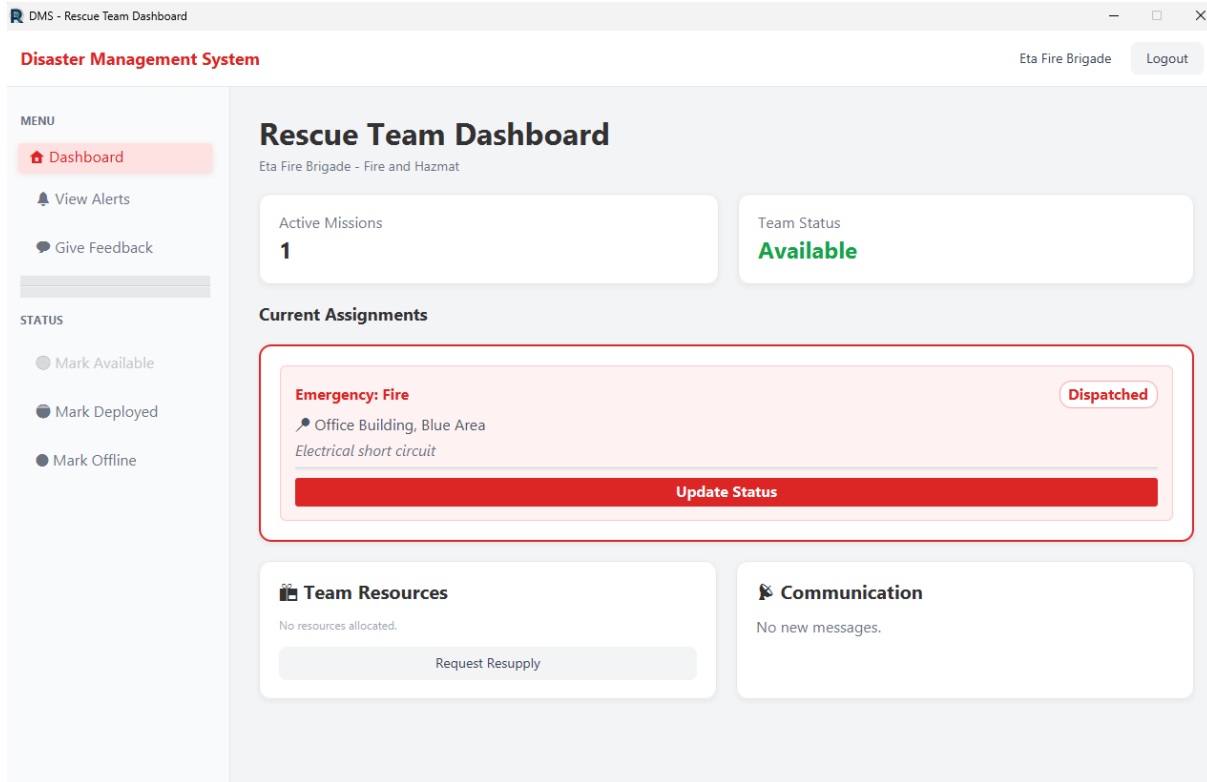
Rawal Dam Flood Relief

Target: Rawal Town, Bhara Kahu

Date: Nov 15, 2025

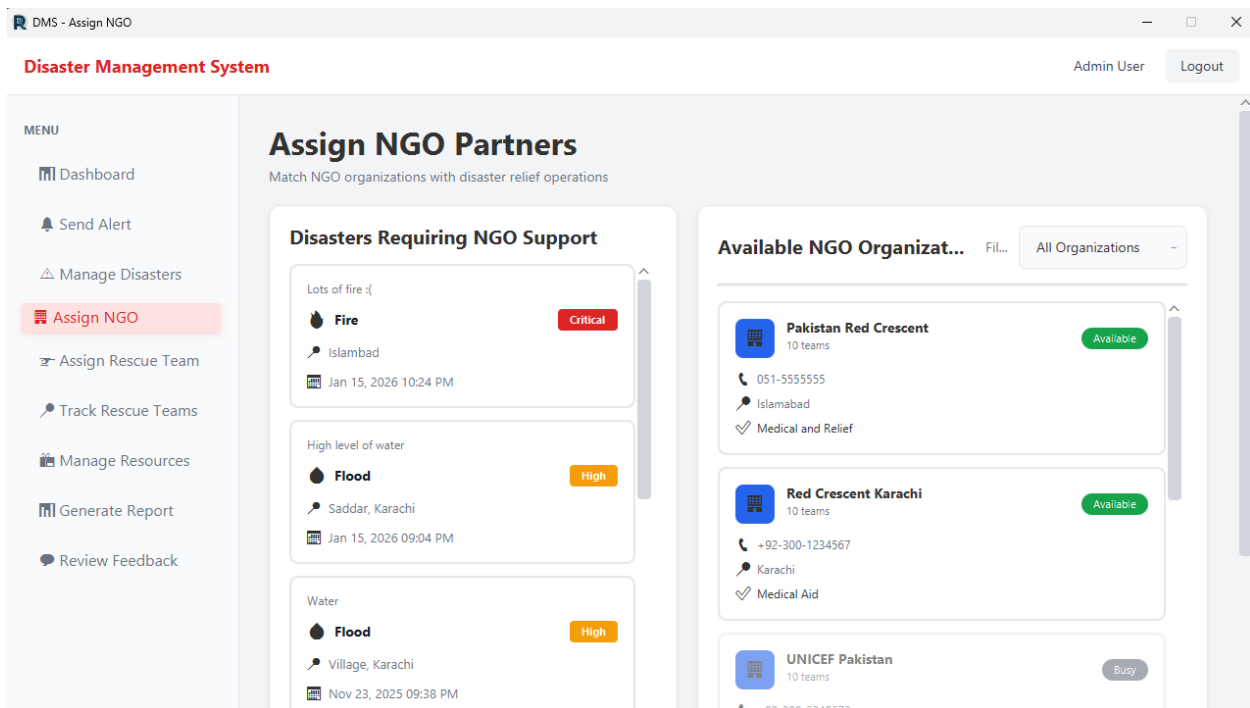
In Progress

8.6 Rescue Team Dashboard

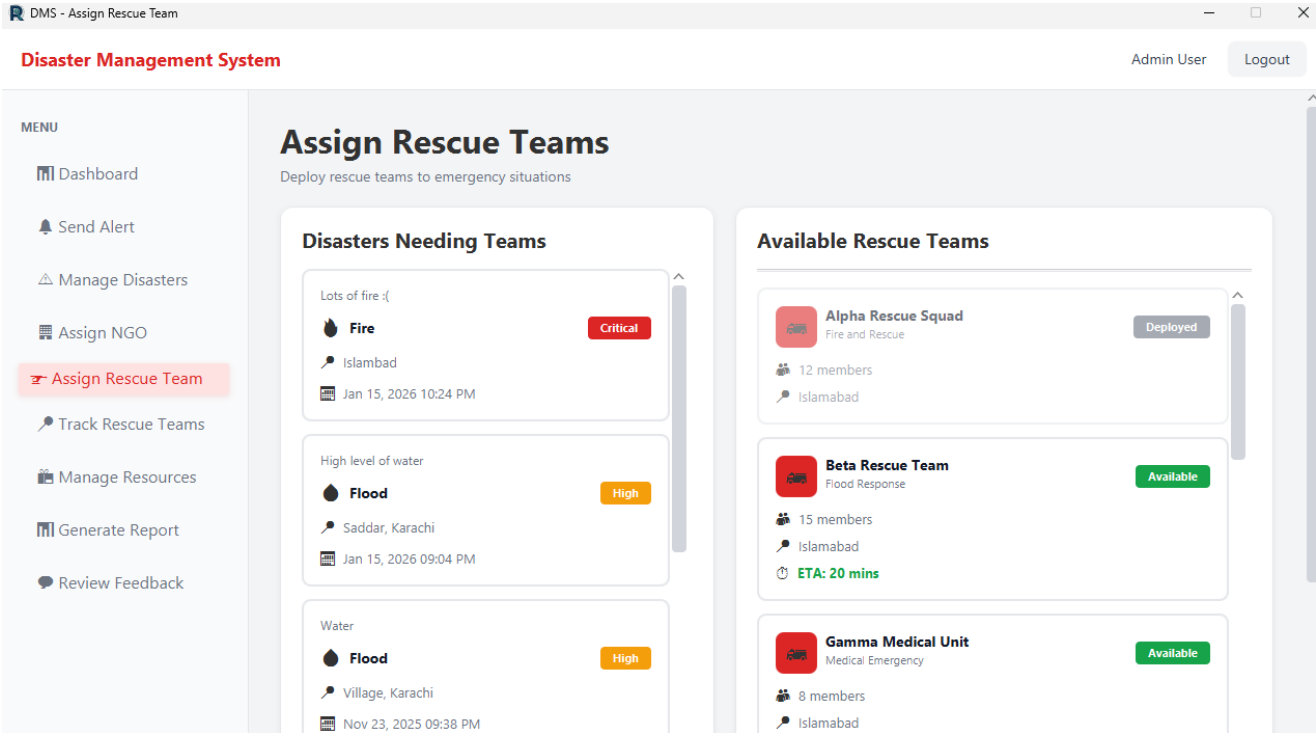


Sprint 3:

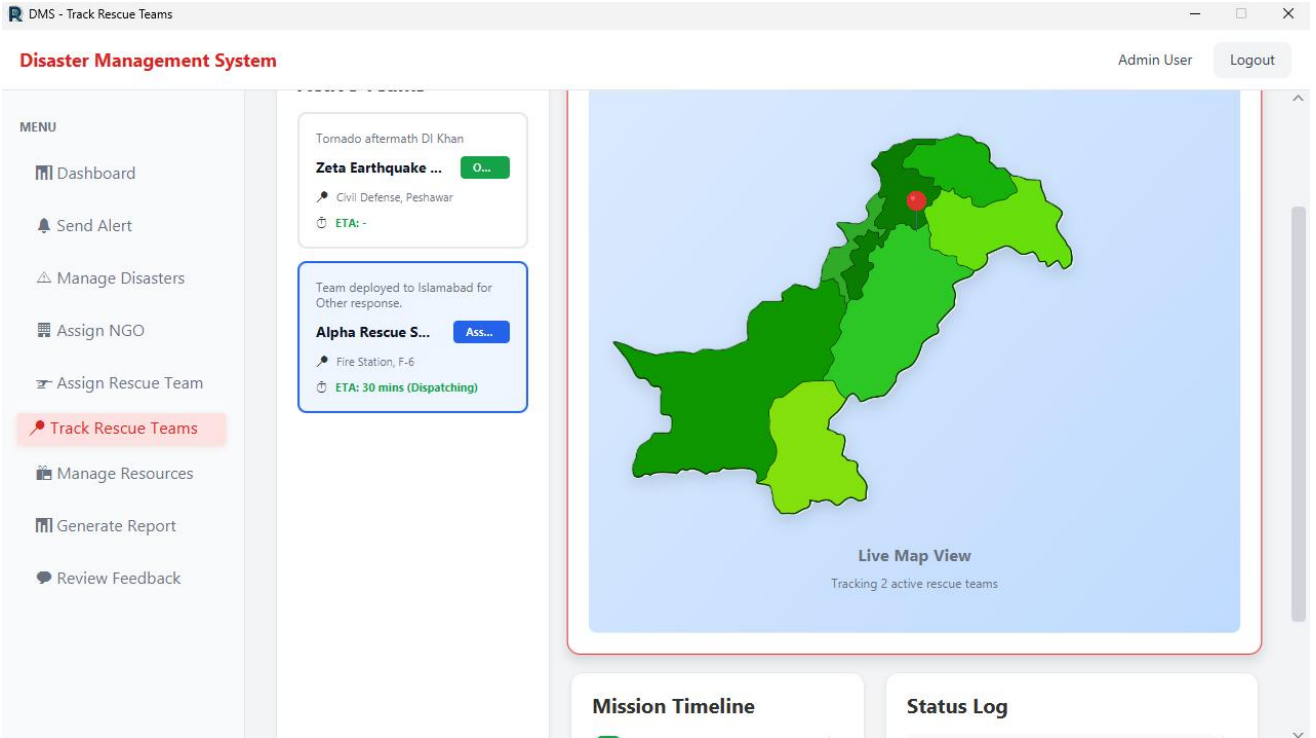
8.7 Assign NGO Page



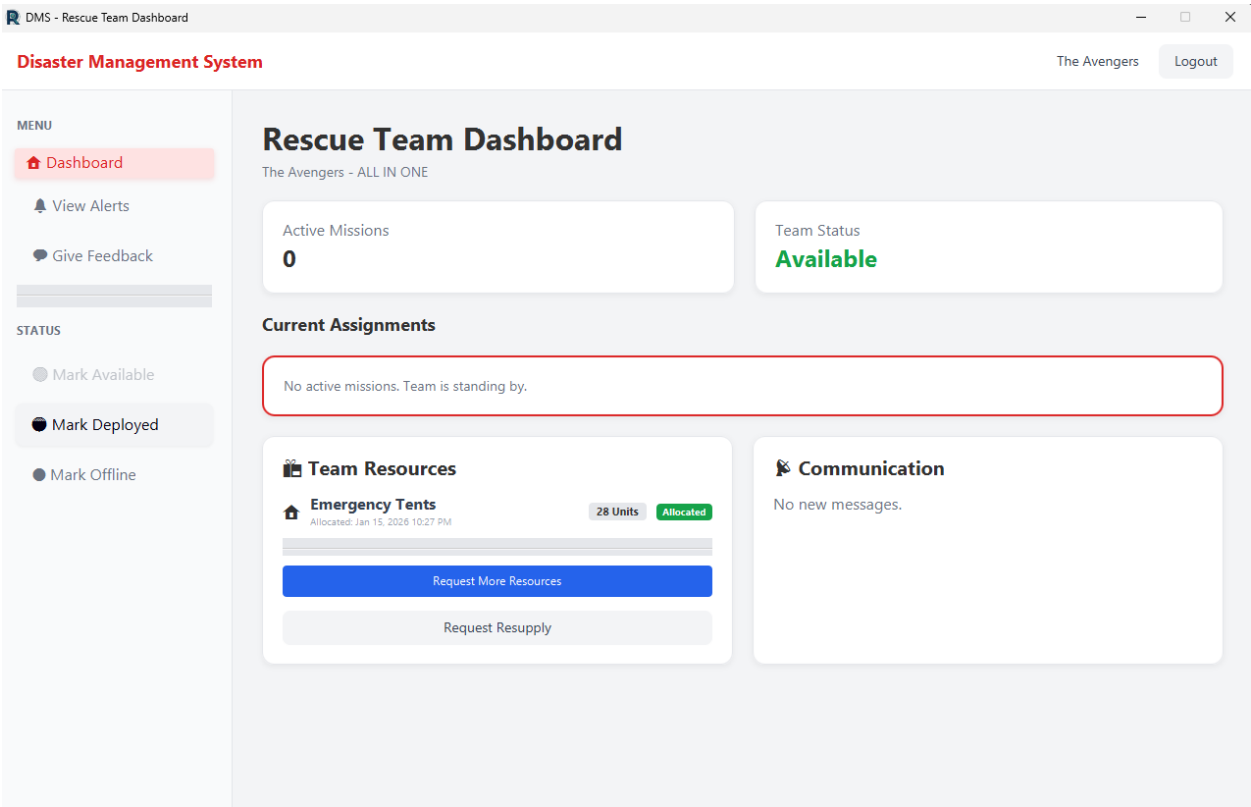
8.8 Assign Rescue Team Page



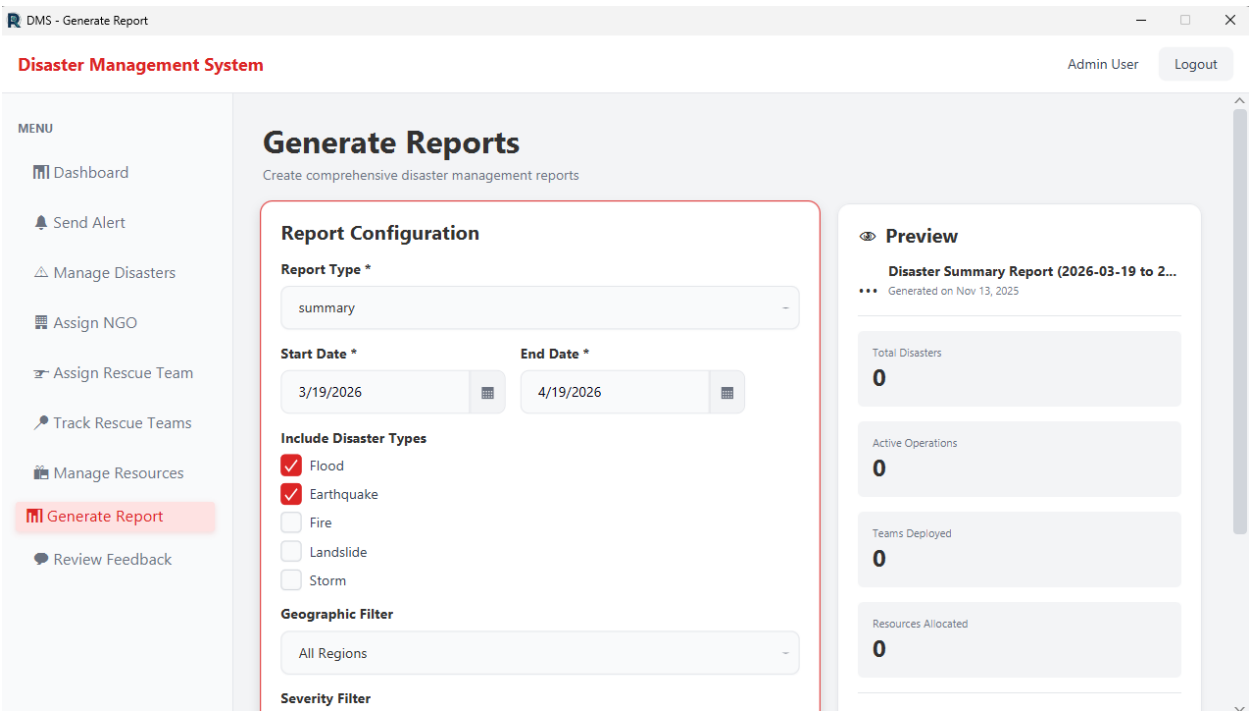
8.9 Track Rescue Teams Page



8.10 Update Mission Status (Rescue Team Dashboard)



8.11 Generate Report Page



8.12 Give Feedback Page

DMS - Feedback

Disaster Management System

Rescue team UserLogout

← Back

Give Feedback

Help us improve our disaster response services

Feedback Details

Feedback Type *

General Feedback

Select Team/NGO (if applicable)

N/A - Not Applicable

Rate Your Experience *

★ ★ ★ ★ ★

★★★★★ Excellent

Subject *

Efficient Resource Delivery

Detailed Feedback *

All requested resources were allocated on time.

8.13 Review Feedback Page (Admin)

DMS - Review Feedback

Admin UserLogout

MENU

- Dashboard
- Send Alert
- Manage Disasters
- Assign NGO
- Assign Rescue Team
- Track Rescue Teams
- Manage Resources
- Generate Report
- Review Feedback

Review Feedback

Manage and respond to citizen feedback

Total Feedback
12

Pending
0

Reviewed
6

Avg Rating
4.1

all-
all-

Feedback Submissions (12)

I LOVE IRONMAN!	Reviewed	RescueTeam	★★★★★
ID: FB012 • Steve Ansari • Apr 18, 2026 • Related to: RT010	<button>View Details</button>	<button>Pending</button>	<button>Resolved</button> <button>Dismissed</button>
Ironman came flying and saved meee!!			
Good team	Reviewed	General	★★★★★
ID: FB011 • Anonymous • Jan 15, 2026 • Related to: RT001	<button>View Details</button>	<button>Pending</button>	<button>Resolved</button> <button>Dismissed</button>

9. Work Division

Sprint 1:

- **Moiz Ansari:** Analyzed and figured out the required user stories. Divided tasks among members. Created team information document and created base document for sprint 1. Created a Trello board. Created github repository. Provided UI design for the system as well.
- **Rehan Tajik:** Created Deliverable Document.
- **Zohaib Umer:** Developed US-01 to US-05 and sub-stories (CRUD operations).
- **Tayyab:** Developed US-06 to US-10 and sub-stories (CRUD operations).

Sprint 2:

- **Moiz Ansari:** Analyzed and figured out the required user stories. Divided tasks among members. Updated documents for sprint 2. Created a Trello board. Developed Manage Resources use case and some other files.
- **Rehan Tajik:** Developed US-16 to US-18 and sub-stories (CRUD operations).
- **Zohaib Umer:** Developed US-11 to US-15 and sub-stories (CRUD operations).
- **Tayyab:** Wrote all structured specifications.

Sprint 3:

Team Member	Tasks / Responsibilities	Deliverable Contribution
Muhammad Moiz Ansari (Team Lead / Scrum Master)	Assigned tasks across the team. Developed Assign NGO & Assign Rescue Team modules (US-19–US-24). Developed Track Rescue Teams with map integration (US-25–US-27). Updated all deliverable documents.	Part B (Sprint Overview, User Stories, Scrum Board setup), Section 5 (Tests) Part C (Screenshots attached) Code final fixes, database
Rehan Tajik	Developed the Update Mission Status module (US-28–US-30). Integrated mission timeline and status log in Track Teams view. Performed black-box testing for Track & Mission modules.	Part B Section 3 (US-19 – US-22) Part C
Zohaib Umer	Developed the Generate Disaster Report module (US-31–US-33). Implemented PDF/Excel/CSV export using iText and Apache POI. Performed black-box testing for the Report module.	Part B Section 3 (US-23 – US-30)
Muhammad Tayyab	Developed Give Feedback & Review Feedback modules (US-34, US-35). Wrote all 4 structured specifications. Prepared test plan and bug report table.	Part B Section 3 (Specs 1-4, US-31 – US-35)